MIL-R-19390A(SHIPS)
28 December 1962
SUPERSEDING
MIL-R-19390(SHIPS)
12 March 1956

MILITARY SPECIFICATION

RADIO SETS AN/SRN-6( )/AN/GRN-9( ) TACTICAL AIR NAVIGATION,

SHORT RANGE

1. SCOPE

1.1 This specification covers the ship or shore based transmitter-receiver used to provide 100 properly equipped aircraft with accurate navigational information within a 200-mile line of sight range. The transmitter and antenna system transmit amplitude and time modulated pulse pairs on one of 126 one megacycle (mc.) channels between 961.5 to 1024.5 and 1150.5 to 1213.5 mc. which will provide a visual indication in the aircraft of the azimuth bearing of the aircraft from the antenna system. The receiver receives pulse pairs transmitted from the aircraft on one of 126 one-mc. channels between 1024.5 and 1150.5 mc. The receiver uses these pulse pairs to trigger the transmitter making it possible for the aircraft to obtain a visual indication of its distance from the antenna system. The design of the system places the burden of space, weight, and complexity on the ship or shore based equipment specified herein. For this specification, hereafter Radio Sets AN/SRN-6( )/AN/GRN-9( ) shall be referred to as transponder beacon equipment.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-I-16910 - Interference Measurement, Radio, Methods and Limits; 14 Kilocycles to 1000 Megacycles.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts: Preparation for Delivery of.
MIL-C-21240 - Crystal Unit Quartz, CR-65/U.

FSC 5825

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STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-415 - Test Points and Test Facilities for Electronic Systems and Associated Equipment, Design Standard for.

PUBLICATIONS

BUREAU OF SHIPS

NAVSHIPS 93820 - Handbook for the Prediction of Shipboard and Shore Electronic Equipment Reliability.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, New York.)

3. REQUIREMENTS

3.1 Preproduction sample.- Prior to beginning production a sample shall be tested as specified in 4.3.1 (see 6.2.1).

3.2 General requirements.- The equipment shall meet the requirements of MIL-E-16400, except as otherwise specified herein. The design objectives, as specified in MIL-E-16400 for reliability, simplicity, ease of installation and maintenance consistent with the needs of the Naval service, shall be considered during all phases of the equipment design.

3.3 Material.- Material shall be in accordance with MIL-E-16400, except as otherwise specified herein. Particular attention shall be paid to the acceptable and unacceptable material requirements of MIL-E-16400.

3.3.1 Temperature.- The equipment shall meet the temperature requirements of class 1 of MIL-E-16400.

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3.3.1.1 Ventilation and cooling.- Ventilation shall be in accordance with MIL-E-16400. Each cabinet shall be equipped with blowers and associated equipment to assure that no part in any unit specified herein shall operate above the temperature specified by the parts manufacturer. Air filters shall be located at all air intakes and shall be readily removable for cleaning without disassembly of the equipment.

3.3.2 Tropicalization.- Tropicalization will not be required.

3.3.3 Test points and test facilities.- Test points and test facilities shall be provided in accordance with MIL-STD-415.

3.3.4 The transponder beacon equipment and accessories shall be designed for unattended operation and when installed and operating, shall radiate a signal which will give meter readings in an aircraft equipped with associated airborne TACAN equipment of bearing and distance from the operating equipment specified herein to the aircraft.

3.3.5 Power source.- The transponder beacon equipment and accessories shall be capable of operation from a primary source of 60 $\pm$ 2 cycles per second (c.p.s.) for continuous operation or 60 $\pm$ 5 c.p.s. for short term random variations. Units shall be capable of satisfactory continuous operation with maximum variations of plus or minus 10 percent in line voltage or voltage of any phase. Variation in the line voltage shall not be synchronous with respect to the rotational frequency of the antenna. If any of the tube specifications call for closer regulation than plus or minus 10 percent, a variac and voltmeter or automatic regulating device shall be used in the primary side of the filament transformer for controlling voltage. Momentary impairment of operation due to a plus or minus 30 percent transient from nominal line voltage of 1-second duration will be permissible, but shall not cause failure of parts or prevent resumption of normal operation when the transient has ceased.

3.3.5.1 Shipboard installation voltages.- For shipboard installation, 440-volt, three-wire, three-phase; and 117-volts, single-phase, will be available. The contractor shall provide a transformer for converting the 440-volt, three-wire, three-phase, to 208-volts, four-wire, three-phase to supply the transmitter-receiver, the antenna group and associated units above and below deck.

3.3.5.2 Shore installation voltage.- For fixed or mobile shore installations all equipment specified herein shall operate from a 208-volt, four-wire, three-phase supply.

3.3.5.3 Primary power circuits.- Primary power circuits shall not be directly grounded. Where capacitive type of grounding is necessary, such capacitance shall be as small as possible. All leads from any primary power supply shall be

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individually protected against damaging overload between the service connection and any other part of the system, equipment, or unit, as applicable.

3.3.5.4 Input and output circuits.- If grounding externally must be confined to specific input and output terminals, these terminals shall be so designated. There shall be no connections between input and output circuits of any regulator. Both input and output shall be isolated from the case.

3.3.5.5 D.C. resistance.- The d.c. resistance between input to output terminals, and from both input and output terminals to case shall be at least 500,000 ohms.

3.3.5.6 Phase unbalance.- The voltage of any one phase of the 3-phase power supply shall not differ from the mean voltage by more than $\pm$ 2 percent.

3.3.5.7 Loss of phase protection.- The protection of circuitry utilizing 3-phase power from the failure of one or more phases shall be provided. This shall consist of circuitry to protect the cooling blowers and circuitry to provide master protection of the whole equipment against loss of input phase or phases, which does not detract from the basis equipment reliability.

3.3.5.8 Klystron aging switch.- A switch, front panel mounted, shall be provided to reduce the primary high voltage supplied during the aging process.

3.3.5.9 Regulation.- Three-phase line voltage regulators if required shall be of the type that regulate each phase line to neutral voltage independently. All performance requirements of this specification shall be met with the voltage regulators in place and the line voltage and frequency variations applying at the input to the voltage regulators. The degree of regulation of all alternating current (a.c.) and D.c. voltages and currents shall be sufficient to insure that no visible change in the indicated readings occur when the power input voltage (including transients) is within the range of 3.3.5. No electron tubes or parts with moving parts shall be used in these regulators.

3.3.6 Operating conditions.- Equipment shall be capable of operation within the accuracies specified herein in accordance with class 1 of MIL-E-16400. Equipment shall also be capable of continuous operation at an altitude of 5000 feet throughout the required temperature and humidity range and with a maximum of 110 percent of the nominal line voltage.

3.3.7 Enclosure.- In accordance with the requirements of MIL-STD-108, the degree of enclosure for all equipment operating below deck shall be dripproof, for all equipment operating above deck shall be spraytight, and for the shore antenna control unit shall be protected. In general, the degree of enclosure for dripproof

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and spraytight equipments will be considered satisfactory if no large amounts of water shall enter or collect in the equipment and the performance shall not be adversely affected by the entrance of this moisture.

3.3.8 Size.- The size of each cabinet shall be 72 inches high, 25 inches wide, and 34 1/8 inches deep.

3.3.9 Shock, vibration and inclination.- All equipment, shall withstand class A shock tests of MIL-E-16400. The contractor shall repair any and all damages resulting from the tests.

3.3.10 Cable entrances.- Power and control cables shall enter the transponder beacon equipment cabinets through cable entrance boxes located on the rear of each cabinet near the bottom. Removable blank entrance plates shall be provided on all four sides of the cable entrance boxes. These plates shall be large enough to permit all power and control cables for any cabinet to enter the cable entrance box through stuffing tubes bolted or welded to the drilled entrance plates. Video or low level r.f. cables shall enter the cabinets through the cable entrance boxes by means of series C r.f. connectors. Transmitter r.f. power shall be fed from an r.f. cable connector on the front of the transmitter-receiver group at either side. Entrance plates for stuffing tubes and r.f. cable connectors shall be provided for other groups and units as necessary. Associated r.f. cable jacks and plugs shall be furnished and shall be capable of clamping armor.

3.3.11 Convenience outlets and telephone jacks.- A convenience outlet properly fused shall be provided in the transmitter-receiver group for connecting a soldering iron or external test equipment or both. A telephone jack, to permit connection of a sound-powered telephone shall also be provided in the transmitter-receiver group and shall be wired to the terminal board. All groups or units which are not to be located in the immediate vicinity of the transmitter-receiver group shall be similarly provided with convenience outlets and telephone jacks.

3.3.12 Elapsed time meters.- Elapsed time meters shall be provided in the equipment to indicate "filament hours" and "h.v. plate hours".

3.3.13 Control knobs.- Control knobs shall be in accordance with MIL-E-16400 except that tactile identification is not required.

3.3.14 Control locking devices.- Control locking devices shall be furnished in accordance with MIL-E-16400. This requirement does not apply to knobs affixed to tap switches or other controls having detent mechanisms incorporated therein.

3.3.15 Blown fuse indicators.- Blown fuse indicators shall be provided in this equipment.

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3.3.16 Remote operation.- Control circuits shall be designed for remote standby, remote start, and remote stop operations. Standby shall be defined as available for operation at 80 percent of full power output in less than 10 seconds.

3.3.17 Automatic restoration.- Operation of transponder beacon equipment shall be automatically restored to full operation after restoration of primary power in cases of power failure. Operation shall be restored as rapidly as possible, consistent with requirement for protection of units. In the event that an overload relay is tripped, the overload shall be automatically reset three times and then shall remain open. If operation is restored and the overload is not tripped again for a period of 20 seconds, the automatic reset shall be recycled to manual position. Instantaneous shutdown of equipment is permissible under class A shock test (see 3.3.9); however; operation shall be immediately restored upon completion of shock transients.

3.3.18 Finish.- Finish of the equipment shall be as specified in MIL-E-16400.

3.3.19 Corrosion protection.- The equipment shall be protected from corrosion in accordance with the requirements of MIL-E-16400.

3.4 Definitions.- For purposes of this specification, the following definitions shall apply:

3.4.1 Duty cycle.- Duty cycle is the ratio of actual energy transmitted in 1 second to the energy which would be transmitted in 1 second if the peak power were transmitted continuously.

3.4.2 Pulse count limiting.- Pulse count limiting is an automatic temporary limitation of equipment replies to protect the transmitter from overload in the presence of excessive traffic (interrogations).

3.4.3 Miles.- The word "miles" as used herein means nautical miles.

3.4.4 Operating channel.- The operating channel is the combination of one interrogation frequency and one reply frequency, constituting a particular navigation operating channel.

3.4.5 Pulse fall time.- Fall time of a pulse is the time required for the trailing edge of a pulse to fall from 90 percent to 10 percent of the maximum voltage amplitude.

3.4.6 Pulse duration.- Pulse duration shall be the time measured between points on the leading and trailing edges of a pulse which are at 50 percent of the maximum voltage amplitude of the pulse.

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3.4.7 Pulse pairs.- The use of a pulse pair in which the constituent pulses of the pair are separated by a specific time interval.

3.4.8 Pulse rise time.- Rise time of a pulse is the time required for the leading edge of a pulse to rise from 10 percent to 90 percent of the maximum voltage amplitude.

3.4.9 Pulse top.- Pulse top is the instantaneous amplitude of the pulse between the point of the leading edge which is 95 percent of the maximum pulse amplitude and the point of the trailing edge which is 95 percent of the maximum pulse amplitude.

3.4.10 Squitter rate.- Squitter rate is the rate of random firing of the transmitter in the absence of interrogations.

3.4.11 Time measurement with pulses.- Measurement of spacing between pulses and other time measurements made with pulses as a reference shall be made with reference to the instant when the leading edge of the pulse reaches 50 percent of its maximum amplitude.

3.4.12 Triggering level.- Triggering level is the sensitivity level of the receiver at which a signal will cause the transmitter to fire 60 percent of the time in the absence of reference and identity pulses.

3.5 Detail requirements.-

3.5.1 Equipment composition.- Equipment shall include a transmitter-receiver group and power supply-test set group, transformers for shipboard installations, and two crystal units each for the transmitter.

3.5.1.1 The transmitter-receiver group shall consist of the following parts:

(a) Coder-indicator.
(b) Radio receiver.
(c) Amplifier modulator.
(d) Frequency multiplier oscillator.
(e) Control duplexer.
(f) Power supply.
(g) Accessories.

3.5.2 Equipment performance.- The equipment shall perform under the following operating conditions:

3.5.2.1 Distance reply signal.- The distance reply signal shall consist of a pair of pulses transmitted in response to a pair of interrogating pulses. The time

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delay in the ground transponder measured from the leading edge of the second interrogating pulse to the leading edge of the second reply pulse shall be 50 ± 0.25 microseconds (μsec.).

3.5.2.1.1 The transponder shall reply with not more than 30 percent countdown to 3300 interrogations per second.

3.5.2.2 Traffic capacity.- The equipment shall provide distance measuring service to 100 interrogating aircraft.

3.5.2.3 Operating channels.- The equipment shall be capable of providing 126 operating channels. The nominal interrogation frequency and the nominal reply frequency for all operating channels shall have a frequency separation of 63 mc.

3.5.3 Pulse shape.- The pulse envelope as detected by a linear detector shall have a rounded shape falling within the following limits:

3.5.3.1 Pulse top.- The instantaneous amplitude of the pulse shall not fall below a line which is 95 percent of the maximum voltage amplitude of the pulse and is parallel to the base line.

3.5.3.2 Pulse rise time.- The pulse rise time shall be 2.5 ± 0.5 μsec.

3.5.3.3 Pulse fall time.- The pulse fall time shall be 2.5 ± 0.5 μsec.

3.5.3.4 Pulse duration.- The pulse duration shall be 3.5 ± 0.5 μsec.

3.5.4 R.f pulse spectrum.- The energy level contained in a 0.5 mc. band centered on a frequency plus or minus 0.8 mc. from the channel frequency, shall be at least 60 decibels (db) below the energy level contained in a 0.5 mc. band centered on the channel frequency. The energy level contained in a 0.5 mc. band centered on a frequency plus or minus 2.0 mc. from the channel frequency shall be at least 65 db below the energy level contained in a 0.5 mc. band centered on the channel frequency. Each lobe of the spectrum shall be of less amplitude than the adjacent lobe nearer the center frequency. The use of filters external to the main cabinet will be considered if required to meet the requirement herein. No thermal control of the spectrum shall be required to protect against drift due to humidity and temperature.

3.5.4.1 CW radiation.- CW radiation between pulses shall not exceed 5 microwatts.

3.5.5 Pulse droop.- Prior to amplitude modulation by the antenna of the composite r.f. pulse train, any pulse in a pair, or in the main reference group, shall not deviate from the average peak amplitude by more than plus or minus 1.0

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percent and any pulse in an auxiliary reference group shall not deviate by more than plus or minus 1.0 percent. The purpose of this requirement is to limit 15 cycle and 135 cycle parts in the unmodulated pulse train. These parts induce cyclical error in the airborne equipment in accordance with the following approximate formulae: Error in degrees is approximately equal to two times a constant, times the percent droop, divided by the percent modulation. The constant represents recovery time and is unity for the maximum error condition.

3.5.6 Pulse coding.- The pulses shall be coded in pairs with a spacing which shall be 12 $\pm$ 0.25 usec. as measured at the 50 percent maximum voltage amplitude points from the leading edge of the first pulse to the leading edge of the second pulse.

3.5.7. Transmitter unit.- The transmitter unit shall be of the pulse type and shall employ a hard tube modulator. The transmitter output frequency shall be controlled by direct crystal multiplication.

3.5.7.1 Transmitter power output.- The transmitter peak power output shall be at least 7.5 kilowatts (kw.) at rated line voltage as measured at the transmitter r.f. output connector or if an external filter is required, at the output of the filter.

3.5.7.2 Transmitter duty cycle.- The duty cycle shall be sufficient to meet the requirements of pulse repetition rate specified in 3.5.7.3 increased by 2.5 percent and the requirements of pulse duration actually produced by the transmitter. This shall be accomplished without exceeding electron tube ratings.

3.5.7.2.1 Transmitter pulse count limiting.- From zero load to full load, the pulse count limiting shall be effective to maintain the average number of transmitted pulses constant to within plus or minus 2.5 percent of its nominal value.

3.5.7.3 Transmitter pulse repetition rate.- In addition to the reference pulse groups, the signal shall have a randomly distributed pulse repetition rate which shall be maintained at 2700 $\pm$ 90 pairs per second before the reference groups are added to the signal; the random rate may include distance replies. The distribution of the random pulse spacing for no interrogation load, shall fall within the limits shown on figure 1, for a dead-time of 60 μsec.

3.5.7.4 Main reference pulse group.- The 15-cycle reference bearing signal, called the main reference group, shall consist of a group of 12 pairs of pulses. The spacing between pairs shall be 30 $\pm$ 0.3 μsec. The repetition rate of the group shall be 15 c.p.s. $\pm$ 0.2 percent.

3.5.7.5 Auxiliary reference pulse group.- The 135-cycle reference bearing signal, called auxiliary reference group, shall consist of a group of six pairs of

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pulses. The spacing between pairs shall be 24 ± 0.25 μsec. The repetition rate of the group shall be 135 c.p.s. and shall be precisely synchronized with the main reference group; however, the auxiliary group which would otherwise coincide in time with the main reference group, shall be removed so that only the main reference group appears in this position in the final output signal.

3.5.7.5.1 The main and auxiliary reference groups shall have precedence over random, identity and distance reply pulsed and these pulses shall not appear during a main or auxiliary reference group.

3.5.7.6 Transmitter frequency.- The transmitter shall be capable of being tuned to any one of 126 one-mc. channels between 961.5 to 1024.5 and 1150.5 to 1213.5 mc. The mid-frequencies of the limiting channels shall be 962, 1024, 1151, and 1213 mc. Not more than 30 minutes shall be required to change crystals and realign tuned circuits in switching channels, when prescribed special test equipment is available.

3.5.7.7 Transmitter crystal oscillator.- A crystal unit in accordance with MIL-C-21240 shall be used. The crystal oscillator circuit shall be approved by the bureau or agency concerned. The crystal oscillator frequency shall be within plus or minus 0.001 percent of the selected channel frequency. Crystal units furnished with each equipment shall be for one frequency but the frequencies will be different for other equipments. Upon request from the contractor, the bureau or agency concerned will supply crystal frequencies 90 days in advance of scheduled completion of equipment assembly.

3.5.7.8 Transmitter frequency stability.- The frequency shall not deviate from the nominal frequency by more than ± .001 percent under all service conditions specified.

3.5.7.9 Frequency multipliers.- Crystal frequency multiplier stages, for the high and low band, may be either fixed tuned or tunable over the frequency range of the band, as required by other design considerations. If tunable circuits are used, there shall be no ambiguity in obtaining proper tuning. Resonance for tunable circuits shall be indicated by at least a 20 percent change in the measured parameter. There shall be no evidence of parasitics or other spurious oscillations in any stage while tuning over the full range with normal excitation to each stage or with the crystal removed.

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3.5.7.10 Transmitter identification signal.- The identification signal shall consist of a series of paired pulses transmitted at a repetition rate of 1350 pulse pairs per second, phase-locked within ≤ 50 μs of the tenth harmonic of the 135 cycle reference bearing signal. The first pulse of an identity pair shall occur 740 ≤ 50 μsec. after the first pulse of each auxiliary reference group.

3.5.7.10.1 Equalizing pulse pair.- To preserve a constant duty cycle and to eliminate bearing error during identity signals, an equalizing pair of pulses shall be transmitted 100 ≤ 10 μsec. averaged over a period of 1 second after each identity pair.

3.5.7.10.2 Identity cycle.- The random and distance reply pulses shall be replaced by the identity pulses at a recurrence rate of once every 37.5 seconds. During identity, the beacon code shall be transmitted as dots and dashes (Morse code) of identity pulses; the spaces between dots and dashes being occupied by random pulses.

3.5.7.10.3 Transmitter identification keyer.- An identification keyer capable of keying the transmitter with three letters and a numeral shall be provided. The keyer shall consist of a keyer cam wheel driven at a nonsynchronous speed of 8 revolutions per minute (r.p.m.) and a follower-and-contact assembly capable of keying the transmitter. Only every tenth revolution of the keyer cam wheel shall be used to key the transmitter. The setting up of any specified code or timing adjustment shall be accomplished by one man within 15 minutes. The motor shall be a Holtzer Cabot Type RWC-2505, or equivlant, conforming to MIL-I-16910. The motor shall have ball bearings and corrosion-resisting steel shafts.

3.5.7.10.3.1 The keying cam wheel shall have 60 adjustable heat-treated steel cam segments, each comprising 6 degrees of arc. This arrangement shall allow the setting up of code characters and, in combination with the follower-and-contact assembly, shall provide the specified character timing, space adjustment, and code changing.

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3.5.7.10.3.2 The follower-and-contact assembly shall consist of a roller and lever operated switch with contacts rated at a 0.5 ampere minimum. Adjustment shall be provided for reestablishing the code character and element timing when, due to wear, the dots, dashes, and spaces depart from nominal timing by amounts up to 50 percent.

3.5.7.10.3.3 When the keyer motor is running under normal operating conditions, and with optimum follower-and-contact assembly adjustment, code character and element timing shall be as specified hereinafter, within a tolerance of plus or minus 20 percent:

Dot length	0.125 second
Dash length	Three dots
Space between dots and dashes within a character	One dot
Space between two consecutive characters of a code group	Three dots

3.5.8 Receiver.- The receiver shall consist of a sensitive superheterodyne type, a passive TR switch, and a decoder designed to process distance interrogation pulses and trigger the transmitter. The receiver shall use the CW output from the transmitter frequency-determining oscillator multiplier chain as the local oscillator signal. A low noise type mixer and an amplifier providing an intermediate frequency of 63.0 mcs. shall be provided.

3.5.8.1 Receiver frequency.- The receiver shall be capable of being tuned to any one of 126 one-mc. channels between 1024.5 and 1150.5 mc. The mid-frequencies of the limiting channels shall be 1025 and 1150 mc. Not more than 15 minutes shall be required to realign tuned circuits when switching channels.

3.5.8.2 Receiver frequency stability.- The frequency of the receiver local oscillator signal shall be determined by the transmitter frequency crystal controlled oscillator and multiplier circuits and shall be maintained within ± 0.001 percent of the specified channel reply frequency. The center frequency of the receiver pass band shall not deviate from the specified interrogation channel frequency by more than ± 50 kilocycles (kc) under service conditions.

3.5.8.3 Receiver triggering level.- A properly coded interrogation 125 db below 1 watt measured at the input connector on the receiver chassis shall be sufficient to trigger the transmitter under a condition of no load. A properly coded interrogation 124 db below 1 watt measured at the same point shall be sufficient to trigger the transmitter under a condition of full load. The threshhold signal level shall not be reduced by more than 2.0 db under a condition of full load. The total number of pulses out of the receiver shall be 2700 ± 90 pulses for conditions of normal loading or for a 50 percent overload.

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3.5.8.4 Receiver bandwidth.- The bandwidth of the receiver shall be such that the triggering level shall not deteriorate by more than 3 db when its total frequency drift is added directly to an incoming frequency drift of plus or minus 100 kc.

3.5.8.5 Receiver selectivity.- The receiver shall reject properly coded interrogations centered on frequencies ± 90 kc. removed from the nominal frequency having an amplitude 80 db above the threshold level of a properly coded interrogation signal in the pass band. Signals arriving at intermediate frequencies shall be surpressed 80 db. All other spurious responses within the 960 to 1215 mc. band shall be surpressed 75 db.

3.5.8.5.1 Low-pass filters shall be provided for the suppression of radar interference from 1650 to 10,500 mc. The suppression shall be at least 60 db.

3.5.8.5.2 Image rejection.- Image responses within the 960 to 1215 mc. band shall be surpressed at least 80 db.

3.5.8.5.3 Receiver decoding.- The receiver shall not decode pulse pairs more than 20 percent of the time for any signal level up to 75 db below 1 watt if the pulse spacing is changed by ± 4 μsec. from nominal.

3.5.8.5.4 The sensitivity of the receiver shall not decrease more than 3 db when receiving a pulse pair spaced ± 0.5 μsec. from nominal.

3.5.8.6 Interference suppression.- Circuitry shall be provided in the receiver to reduce its vulnerability to the types of interference specified below.

3.5.8.6.1 Narrow pulse interference.- The receiver shall be unresponsive to pulses having a duration of 1.2 μsec. or less and as strong as 10 dbw. a paired spacing within the tolerances of the decoder, a repetition rate equal to, or lessthan 2500 pulse pairs per second, and at a signal strength of 0 to 130 dbw.

3.5.8.6.2 CW interference.- In the presence of CW interference, the receiver shall be capable of responding to interrogating signals that exceed the interference by 3 db or more. The receiver shall continue to generate randomly distributed pulses at a rate of 2700 ± 150 per second for any and all conditions of CW jamming.

3.5.8.7 Receiver blanking, resonance and automatic gain control (AGC) capture.- Short high-power spot jamming and strong jamming pulses transmitted at random are potentially capable of increasing the dead time in the receiver and may produce, capture or cause instability in the normal and fast AGC circuits. Provisions shall be incorporated to overcome these weaknesses.

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3.5.8.8 Receiver echo suppression.- Properly spaced pulse pairs shall not trigger the transmitter more than 20 percent of the time for any signal 70 db below 1 watt or stronger if the amplitude of the second pulse of the pair is at least 25 db below the amplitude of the first pulse. For interrogation levels greater than 40 dbm., the echo suppression circuits shall not delay the second pulse in a pair with respect to the first pulse of the pair.

3.5.8.9 Receiver recovery time.- If blanking is necessary, the dead time shall be adjustable from 16 to 16 microseconds.

3.5.8.10 Receiver output.- In addition to the video output feeding directly to the transmitted modulator, a video output jack shall be provided on the receiver so that air-to-ground functions which may be added in the future can be abstracted from the receiver and fed to supplementary equipment.

3.5.8.11 Response delay.- The response delay shall be 50 $\pm$ 0.25 μsec. measured from the leading edge of the second interrogating pulse to the leading edge of the second reply pulse measured at a standard signal strength of 50 db above threshold.

3.5.8.12 Duplexer.- The duplexer shall be of a passive type permitting simultaneous operation of the receiver and transmitter in accordance with the performance requirements. Duplexer control shall be calibrated for the entire tuning range.

3.5.8.13 Test and maintenance provisions.- Adequate controls, directional couplers, delay lines, and all other provisions shall be made to permit adjustment of the operating conditions and accurate determination of pulse characteristics in accordance with specification requirements by means of external test equipment (TACAN special test equipment) through test points.

3.5.8.13.1 Signal measurement and observation of wave-form.- Provisions shall be made for observation of wave-form and measurements of incident and reflected power/voltage by installing couplers such as follows:

(a) Input to tripler/output of final doubler.
(b) Input to final amplifier/output of tripler.
(c) Input to spectrum filter/output of final amplifier.
(d) Input to antenna transfer switch/output of spectrum filter.
(e) Input to antenna/output of antenna transfer switch. Monitoring facility shall also be provided between the spectrum filter and antenna.

Note: (d) and (e) shall be bi-directional couplers with an accuracy of $\pm$ 1 db and 35 db directivity.

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3.5.8.14 Power supply.- The power supply assemblies shall be capable of delivering all power requirements for the transmitter-receiver group including the final power amplifier incorporated in the equipment. The power supply assemblies shall also meet the requirements of 3.3.5.

3.6 Accessories.- The following accessories shall be provided:

(a) Adapters and probes-video and r.f.
(b) Cables - r.f. and video.
(c) Calibration charts.
(d) Connectors, plugs, and tee connectors.
(e) Technical manuals. The technical manuals shall provide check out procedures, alignment procedures, maintenance and preventive maintenance procedures based upon the use of TACAN special test equipment.

3.7 Reliability.- The following procedure shall be applied:

(a) The Mean-Time-Between-Failure (MTBF) of the equipment shall be predicted using Method D of publication NAVSHIPS 93820.
(b) The design shall be reviewed and analyzed in detail by a group provided by the contractor independent of the designers to determine that it is inherently as reliable as is practical. It shall be the particular function of this group to constructively criticize such common weaknesses as unnecessary complexity, misapplication of parts, and those commonly called "marginal design". This group shall report the results of the design review and analysis together with recommendations to the bureau or agency concerned and the designers.
(c) Any failure of a prototype, preproduction or production equipment that occurs during the development, construction, or testing of the equipment shall be analyzed and reported to the bureau or agency concerned. The analysis shall be conducted in such a manner as to determine the cause of the failure so that its recurrence can be prevented. Reports of the failures and their analyses shall be forwarded to the design review group for endorsement.

The equipment shall be considered reliable whenever the reliability prediction, the design review, and the failure analyses are completed and the bureau or agency concerned is satisfied that any faults revealed by these studies have been corrected.

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3.8 Workmanship.- Workmanship shall be as specified in MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General inspection.- Unless otherwise specified herein, general inspection shall be in accordance with MIL-E-16400.

4.2.1 Classification of inspection.- The methods of examination and testing of the equipment shall fall within the following classifications:

(a) Preproduction inspection.
(b) Quality conformance inspection.

(1) Production inspection.
(2) Production control inspection.
(3) Environmental tests.

4.2.2 Tests and test procedures.- The contractor shall submit a list of proposed tests and test procedures to the bureau or agency concerned for approval 60 days prior to setting up test equipment for preproduction and quality conformance inspection.

4.3 Preproduction inspection.- Preproduction inspection shall consist of the examination and tests specified in 4.3.1.

4.3.1 Preproduction inspection shall be conducted at the place of manufacture on the prototype model and shall be in accordance with MIL-E-16400, except that shock, vibration, and inclination tests may be performed at a laboratory satisfactory to the bureau or agency concerned if the necessary equipment is not available at the place of manufacture.

4.4 Quality conformance inspection.-

4.4.1 Production inspection.- Production inspection shall be made on every equipment offered for delivery and shall be in accordance with MIL-E-16400. Production inspection shall include the heat test of MIL-E-16400 except that the equipment shall be subjected to a heat test of 48 hours in lieu of 8 hours.

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4.4.2 Production control inspection.- Production control inspection shall be in accordance with MIL-E-16400 and as specified in 4.4.2.1 and 4.4.2.2.

4.4.2.1 Sampling plan.- A random sample of units shall be selected from each lot that has passed the production inspection and shall be subjected to production control inspection. The number selected and lot acceptance level shall be as shown in table I.

Table I - Sampling for production control inspection.

Lot size	Sample size	Number of units nonconforming in any production control inspection test	
		Acceptance inspection	Rejection number
2 to 10	2	0	1
11 to 19	3	0	1
20 to 33	5	0	1
34 to 50	7	0	1
51 to 75	10	0	1
76 to 116	15	0	1
117 to 200	25	1	2

4.4.2.2 Each of the sample units selected in accordance with table I shall be subjected to the production control inspection of MIL-E-16400 except as follows:

(a) Radio interference and undesired radiation tests shall be performed on one out of every 20 equipments produced.
(b) The equipment shall be subjected to a heat test of 48 hours in lieu of 8 hours as specified in MIL-E-16400.

4.4.2.2.1 The results of each test shall be compared with the requirements of this specification. Failure to conform to this specification for any test shall be counted as a defect, and the unit shall be rejected. If the number of rejections in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.4.3 Environmental tests.- Environmental tests shall be conducted in accordance with MIL-E-16400.

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4.4.3.1 Sampling plan.- Random samples shall be selected for environmental testing from each lot which has passed production inspection. The environmental tests may be carried out concurrently with production control inspection on different samples from the lot being tested. The sample size and lot acceptance level shall be as shown in table II.

Table II - Sampling for environmental tests.

Lot size	Sample size	Number of units non-conforming in any environmental test	
		Acceptance number	Rejection number
1 to 35	1	0	1
36 to 100	2	0	1
101 to 300	3	1	2

4.4.3.2 Failure to conform to this specification for any test shall be counted as a defect and the unit shall be rejected. If the number of rejections in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation.-

5.1.1 Basic equipment or item.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

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5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate, and will afford protection against physcial or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.1).

5.2.1.1 Preservation, packaging, packing and marking.- The equipment, accessories, technical publications, equipment and onboard repair parts shall be preserved and packaged by level A or C; packed by level A or B and marked by levels A and B as specified (see 6.1) in accordance with MIL-E-17555.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preservation, packaging, packing or marking requirements other than those required by paragraph 5.1 (see 5.2).

6.2 Preproduction.-

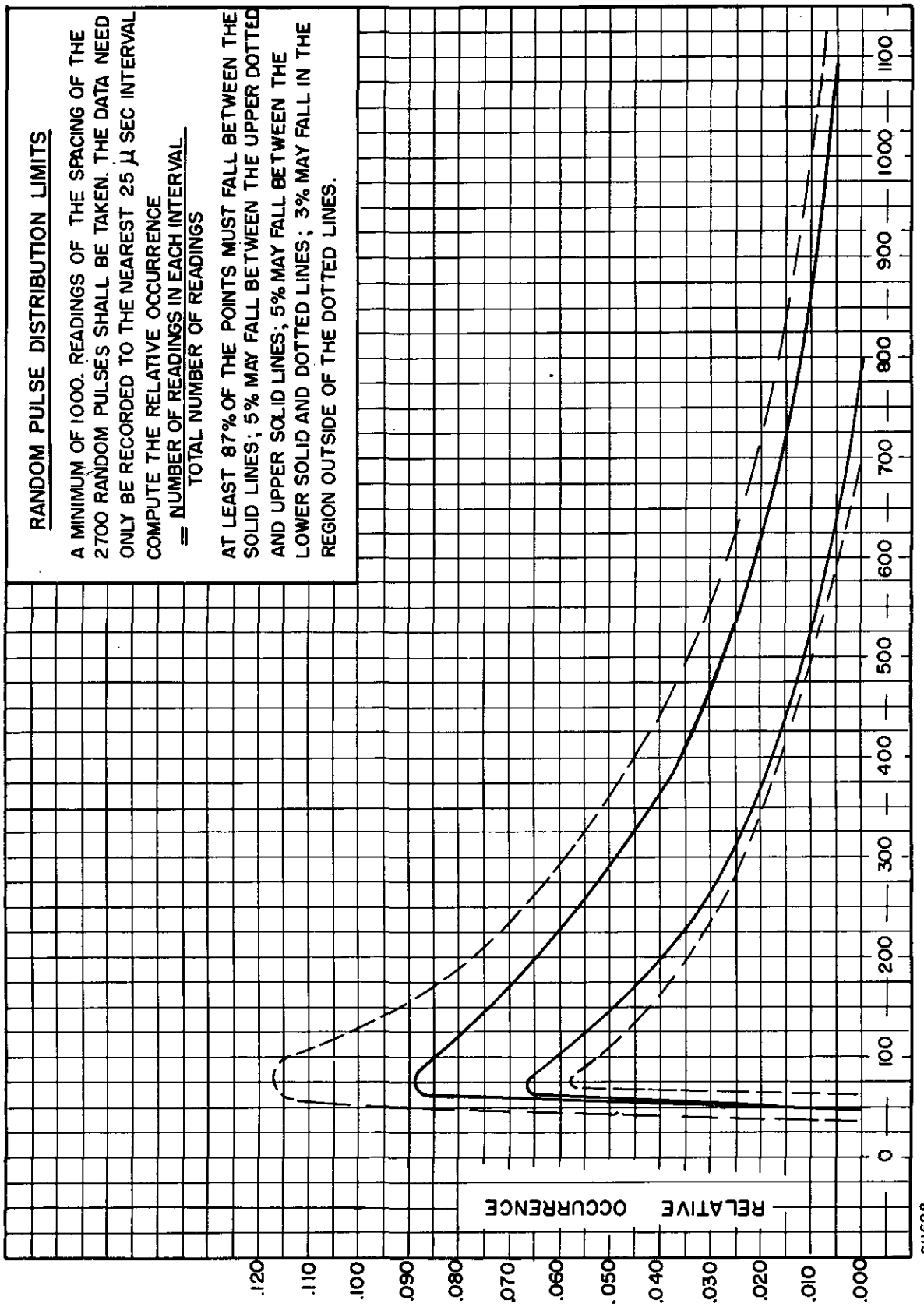
6.2.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procurred or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

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Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

INSTRUCTIONS

BUDGET BU. NO. 45-R309

This sheet is to be filled out by personnel either [illegible]ernment or contractor, involved in the use of [illegible]e specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

[illegible]TTED BY *(Print name and activity)*	DATE

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